


## Carried object

Moving in the world frame: 4 persons, **the handbag**. Staying in place: 6 paintings, the window, the chair.

**The handbag** moves about 0.9 m in the world frame at an average speed of 0.59 m/s.

### Question

**Right answer: A**

In the world frame (ignoring camera motion), approximately how far (in meters) did **the handbag** move between frame 9 and frame 23 of 24?

**A. 0.58** B. 1.4 C. 1.1 D. 0.78

ground truth: 0.62 m

### Qwen3-VL-8B · frames only

```
<evidence>
object: the handbag
position_frame_9_m: (0.8, 1.5, 5.6)
position_frame_23_m: (1.0, 1.6, 6.9)
displacement_m: 1.3
</evidence>
<answer>B</answer>
```

✗

### Ours · frames + trajectory tokens

```
<evidence>
object: the handbag
position_frame_9_m: (1.0, 1.5, 6.5)
position_frame_23_m: (1.2, 1.5, 7.1)
displacement_m: 0.6
</evidence>
<answer>A</answer>
```

✓